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Department of Computer Science & Engineering



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On

A Consensus-Based Family Document Graph Approach for Civil Registry Inconsistency Detection and Correction in Bangladesh

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Abstract

In Bangladesh, civil documents such as NID, Birth Certificate, and Academic Certificate often contain different versions of the same person's name, date of birth, and address. These differences create problems during government correction applications because the current process mostly handles one correction at a time and depends on manual officer checking. This work proposes a consensus-based Family Document Graph approach for finding document mismatches inside a family. The system collects all written versions of a person's name from NID, Birth Certificate, Academic Certificate, and related family documents. It then uses a counting method to select the most common version, separates small name mistakes from serious mismatches, checks date formats, checks parent-child age gaps, and suggests fixed spellings for address parts such as Division, District, Upazila, Union, City Corporation, and Pourosova. A test dataset of 50 to 100 families will be collected in CSV files, and the proposed method will be compared with the old manual correction process using Accuracy, Precision, Recall, and F1-Score.

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Introduction

In Bangladesh, civil documents like the National Identity Card (NID), Birth Certificate, and Academic Certificate are very important for every person. A common problem is that the same person's name, date of birth, or address is written in different ways in different documents. This causes problems in government offices, job applications, passport work, and legal matters.

The problem gets worse in a family. A child's documents must show the same parent names that are in the parents' own documents. If a child's Father Name field does not match the father's NID name, it takes a long time to fix at the office.

Current systems mainly handle one correction application at a time. A user submits related documents, and an officer checks them manually. They do not automatically look at how one person's name appears in all the family's documents. This project gives a new idea called the **Family Document Graph** with a **Consensus Method**. The system finds every place where a person's name is written in all family documents, counts which version is written most, and uses that as the correct value. It also checks Date of Birth, parent-child age gap, and Address for all family members.

Problem Statement

These are the common problems found in Bangladesh documents:

- **Name Mismatch:** "Mohammad Ali" in Birth Certificate but "Md. Ali" in NID. These may be the same person with a small spelling difference, but current systems cannot decide this on their own.
- **Family Name Problem:** A father's name may be written in different ways in his own NID, his son's Birth Certificate, and his daughter's NID. No current system collects all these values together to find the right version.
- **Date Format Difference:** "01/05/1995" and "1995-05-01" are the same date but written in different ways.
- **Wrong Age Gap:** A child must be younger than their parents by a proper number of years.
- **Address Spelling Problem:** The same village or area name may be written in different ways. Division, District, Upazila, Union, City Corporation, and Pourosova names should follow one fixed spelling list.
- **Serious Name Mismatch:** "Mohammad Ali" vs. "Md. Akib" may be two different people. This needs a senior officer to check.

Right now, there is no automatic system in Bangladesh that collects a person's name from all family documents and uses a counting method to find the most correct value.

Literature Review

Civil Registry and Identity Systems

Civil registry and identity systems are used in many countries to store personal identity data. Systems such as NADRA in Pakistan, Aadhaar in India, and CRVS systems in different coun-

tries focus on identity registration, unique identity numbers, birth records, death records, or biometric data [1, 2, 3]. These systems are useful for identity management, but they are usually not designed to collect every family member's NID, Birth Certificate, and Academic Certificate together for family-level document checking.

In Bangladesh, NID and Birth Registration systems allow users to apply for verification or correction. The user submits related documents, and an officer checks the application manually. This process supports correction, but it does not automatically find hidden mismatches in other family members' documents before approval.

Name Matching and Record Linkage

Name matching is a common problem in data cleaning and record linkage. Traditional methods compare strings using edit distance or similarity scores, such as the Levenshtein distance and Jaro-Winkler similarity [4, 5]. These methods can help find spelling differences, but they may not be enough for Bangladeshi name patterns such as "Md." and "Mohammad". A small name difference may still mean the same person, while a large difference may mean a different person.

This project uses a simple name-level decision idea. For example, "Md. Ali" and "Mohammad Ali" can be treated as the same person with a small mistake, but "Mohammad Ali" and "Md. Akib" should be treated as a serious mismatch that needs senior officer checking.

Consensus Methods in Data Cleaning

Consensus-based data cleaning selects a value by checking multiple sources and choosing the value that appears most often or has the strongest support [6]. This idea is useful when the same person's name appears in many documents. Instead of checking only one application, this project collects all name versions from the whole family and uses a counting method to find the most supported value.

Relationship and Address Validation

Some data errors cannot be found by name matching only. Parent-child relationships also need date-of-birth checks. A parent should be older than a child by a proper number of years. Address fields also need fixed spelling rules because the same area name may be written differently. This project uses a fixed location spelling list for Division, District, Upazila, Union, City Corporation, and Pourosova names.

Literature Gap

Existing identity and civil registry systems manage documents and correction applications, but they do not combine family-level document linking, consensus-based name counting, date-of-birth checks, age-gap checks, and address spelling suggestions for Bangladesh civil documents. This gap is the main research focus of this project.

Old Method and Research Gap

Old Method

The old method is the manual government application process. In Bangladesh, a user can apply through government sites for NID, Birth Certificate, or certificate-related correction. In this method, the user submits related documents with the application. Then an officer checks the submitted documents and approves or rejects the application.

This process is useful, but it is mostly manual. If another family member's document has a wrong name, wrong DOB, or mismatch, the user usually has to correct that document first by applying again. The old method does not automatically find all family-level mismatches before the officer checks the application.

System/Method	Country	What It Does
NADRA	Pakistan	Keeps NID data and body scan information.
Aadhaar	India	Gives a unique ID number, but does not check family document problems.
CRVS Systems	Many countries	Keeps birth and death records.
Bangladesh NID Portal	Bangladesh	Takes NID correction or verification applications with related documents. An officer checks and approves manually.
Bangladesh Birth Registration	Bangladesh	Takes Birth Certificate applications or corrections with related documents. An officer checks and approves manually.

What is Missing in the Old Method

1. It depends on application submission and officer checking.
2. It does not compare NID, Birth Certificate, and Academic Certificate at the same time before approval.
3. It may keep family data but does not collect one person's name from all family documents and check them together.
4. If another family member's document has a mismatch, the user has to correct it manually first through a separate process.
5. It does not check name problems in both Bengali and English together.
6. It does not tell the difference between a small name mistake and a serious name mismatch.
7. It does not check the age gap between parents and children across all related documents.
8. It does not use a counting method to find the most correct value.
9. It does not tell the user which value should be fixed.

Research Gap

This project works on these research gaps:

1. **No family-wide name search:** The old method does not find all written versions of one person's name from every family document and use them to find the right value.
2. **Manual correction before approval:** The old method takes applications and related documents, then officers verify them. But if a family member's related document has a mismatch, the user must correct that document manually first.
3. **No counting-based correction:** No current system counts how many times each name version appears in all family documents and uses the most common one as the correct value.
4. **No multi-document check:** Most systems check only one document. They do not compare NID, Birth Certificate, and Academic Certificate together.
5. **No mismatch level:** The old method does not clearly separate small name mistakes from serious mismatches that need officer checking.
6. **No age gap check across documents:** NID systems do not check the age gap between parents and children across documents.
7. **No Bangladesh-specific system:** Bangladesh has its own NID and Birth Certificate format, but there is no automatic system for this.
8. **No fix suggestion:** The old method may find problems, but they do not tell the user which value to fix.

Research Question

This project tries to answer this question:

Does a consensus-based Family Document Graph give better results than the old manual correction process for finding and fixing name problems in Bangladesh civil registry documents?

Objectives

The main goals of this project are:

1. To build a family-linked database that keeps all document data for every family member.
2. To find every written version of one person's name from all family documents and build a Family Document Graph.
3. To use a counting method that picks the most common name version as the correct value.
4. To separate small name mistakes from serious name problems.
5. To compare Date of Birth by the real date, not the format.
6. To check whether parents and children have a proper age gap.
7. To check Address fields using fixed spelling for Division, District, Upazila, Union, City Corporation, and Pourosova.
8. To test the new method against the current manual single-correction application process using Accuracy, Precision, Recall, and F1-Score.

9. To make a clear mismatch report with fix suggestions.

Scope of the Project

- The system will work for one family with Father, Mother, Son(s), and Daughter(s).
- Documents: NID, Birth Certificate, Academic Certificate.
- Fields: Name (Bn & En), Father Name (Bn & En), Mother Name (Bn & En), Date of Birth, Age Gap, Address.
- The system will work with both Bengali and English text.
- Output: Family Document Graph, most common name, mismatch report, and fix suggestions.

System Design and Methodology

This project follows a research method based on problem definition, dataset creation, old-method baseline testing, proposed-method testing, and result comparison.

Problem Formulation

The input of the system is one family record. A family record contains Father, Mother, Son(s), and Daughter(s). Each person can have values from NID, Birth Certificate, and Academic Certificate. Related family fields, such as a child's Father Name or Mother Name, are also used. The data will be collected and stored in CSV files before processing.

The output of the system is:

- the most supported name value for each person,
- a list of small mistakes and serious mismatches,
- DOB and age-gap validation result,
- address spelling suggestions,
- and a final mismatch report.

Family Document Graph Model

The proposed method models one family as a graph. Each family member is a person node. Each document field that contains a person's name is connected to that person node. For example, the father's name can appear in the father's NID, father's Birth Certificate, son's Birth Certificate, son's NID, and daughter's NID. All of these values become evidence for the father's correct name.

Consensus-Based Value Selection

Let $V_p = \{v_1, v_2, \dots, v_n\}$ be all written versions of a person's name p found across the family documents. The system counts how many times each version appears. The most supported value is selected as:

$$CorrectName(p) = \arg \max_{v \in V_p} count(v)$$

After selecting the most supported value, the system checks the other values. Small differences such as “Md.” and “Mohammad” are marked as small mistakes. Very different names are marked as serious mismatches and need senior officer verification.

DOB, Age-Gap, and Address Rules

All DOB values are first converted to one standard date format. Then the system checks whether the real dates are the same. Date format alone does not create a mismatch.

For age-gap checking, the parent must be older than the child by a proper number of years. In this proposal, the minimum valid parent-child age gap is set as 15 years for testing.

For address checking, the system uses a fixed location spelling list for Division, District, Upazila, Union, City Corporation, and Pourosova. If a document uses a different spelling, the system suggests the fixed spelling.

Research Steps

This project follows four clear steps.

Step 1 — Collect the CSV Dataset

There is no public dataset for Bangladesh civil document problems, so this project will collect a **test dataset** in CSV format using realistic data patterns.

- **Size:** 50 to 100 family records.
- **Format:** CSV files will be used for data collection and testing.
- **Structure:** Each family has Father, Mother, and one or more children. Each person has NID, Birth Certificate, and Academic Certificate fields.
- **Problem types included:**
 - Small name mistakes (e.g., “Md. Ali” vs. “Mohammad Ali”)
 - Serious name mismatches (e.g., “Mohammad Ali” vs. “Md. Akib”)
 - Different date formats for the same date
 - Wrong age gap between parent and child
 - Different spelling for the same address area
- **Correct answers:** For each CSV record, the right value will be marked by hand in a ground-truth column so the system result can be checked against it.

Step 2 — Build the Old Method (Single Correction Application)

Before testing the new method, the current **single correction application** process will be built as the baseline. This old method follows the current government application style: the user

applies for one correction at a time, submits related documents, and an officer checks the submitted documents manually.

For example:

- A user applies to correct one NID field.
- A user applies to correct one Birth Certificate field.
- A user submits related documents for that one correction.
- An officer checks the submitted documents and approves or rejects it.

This old method will not build a Family Document Graph. It will not count how many times a name appears across all family documents. If a mismatch is found in another family member's document, the user has to correct that document manually through another application.

This old method will be tested on the same dataset to get baseline results.

Step 3 — Build the New Method (Consensus Family Document Graph)

The new method will be tested on the same dataset. For each person, the system will:

1. Find every place where the person's name is written in all family documents.
2. Count how many times each name version appears.
3. Pick the most common version as the correct value.
4. Mark other versions as small mistakes or serious mismatches.
5. Check Date of Birth after fixing the format.
6. Check the age gap between parents and children.
7. Check address parts using a fixed location spelling list.
8. Give fix suggestions.

Step 4 — Compare Both Methods

Both methods will be checked using these measures:

Measure	Meaning
Accuracy	Out of all checks, how many did the system get right?
Precision	Out of all problems the system found, how many were real problems?
Recall	Out of all real problems in the data, how many did the system find?
F1-Score	A single number that combines Precision and Recall.

The results of both methods will be shown in a table. This will prove whether the new method is better than the old one. This comparison is the main research part of this project.

System Design

System Parts

The system has three main parts:

1. **Data Input** — CSV files containing all family member document data.
2. **Processing** — Family Document Graph builder, counting engine, name checker, date fixer, and age gap checker.
3. **Output** — Mismatch report with fix suggestions.

Database Design

Two tables will be used:

- **Family Table:** Family_ID, Family_Code
- **Person Table:** All document fields with a Role field (Father / Mother / Son / Daughter) linked to Family_ID

Main Idea: Family Document Graph and Counting Method

For each person, the system finds every place where their name appears across all family documents:

Example — Finding Father's name from all documents:

Document / Field	Name Found
Father's NID Name	Md. Ali
Father's Birth Certificate Name	Mohammad Ali
Father's Academic Certificate Name	Md. Ali
Son's Birth Certificate — Father Name	Mohammad Ali
Son's NID — Father Name	Md. Ali
Son's Certificate — Father Name	Md. Ali
Daughter's Birth Certificate — Father Name	Mohammad Ali
Daughter's NID — Father Name	Md. Ali

Count result:

- “Md. Ali” — 5 times → **Correct value**
- “Mohammad Ali” — 3 times → Small name mistake

Fix suggestion: All documents should use “Md. Ali”.

Other Checks

- **Name Level Check** — Small mistakes (short forms, spelling) vs. serious mismatches (completely different names).
- **Date Fix** — All date formats are changed to one standard before checking.
- **Age Gap Check** — Father and mother must be older than the child by a proper number of years.
- **Address Check** — Division, District, Upazila, Union, City Corporation, and Pourosova names are checked against a fixed spelling list. If a spelling does not match, the system suggests the correct fixed spelling.

Address Master List

For address checking, the system will use a fixed spelling list for Bangladesh location names. The list will contain:

- Division
- District
- Upazila
- Union
- City Corporation
- Pourosova

If a document uses a different spelling from the fixed list, the system will show it as an address spelling mismatch and suggest the fixed spelling. For example, if the fixed spelling is “Chattogram” but a document writes “Chittagong”, the system will suggest “Chattogram”.

Check Logic

For One Person

- Find all name values from all family documents for this person.
- Count each version and pick the most common one as the correct value.
- Mark other values as small mistakes or serious mismatches.
- Check Date of Birth after fixing the format.
- Check each address part against the fixed location spelling list.
- Suggest the fixed spelling if Division, District, Upazila, Union, City Corporation, or Pourosova spelling is wrong.

Across Family Members

- Check Child’s Father Name against Father’s correct name.
- Check Child’s Mother Name against Mother’s correct name.

- Check Father's age gap with each child.
- Check Mother's age gap with each child.

Expected Output

The system will make a report like this:

Person / Field	All Values Found	Correct Value	Result
Father — Name (EN)	NID: Md. Ali Birth: Mohammad Ali Cert: Md. Ali Son Birth F: Mohammad Ali Son NID F: Md. Ali Daughter NID F: Md. Ali	Md. Ali (5/8)	Small Mistake
Father — DOB	NID: 1980-05-01 Birth: 01/05/1980	1980-05-01	Match
Age Gap — Father & Son	Father: 1980 Son: 2005	25 years	Valid
Address — District	NID: Chittagong Birth: Chattogram	Chattogram	Spelling Suggestion
Age Gap — Father & Child	Father: 1980 Child: 1982	2 years	Invalid

The report will also show the comparison table between the new method and the old method, with Accuracy, Precision, Recall, and F1-Score.

Tools and Technology

Data Input: CSV files
Frontend: HTML, CSS, JavaScript
Backend: Python (Django / Flask)
Database: MySQL / PostgreSQL
Logic: Python — counting engine, name checker, date fixer, age gap checker, address spelling checker
Testing: Python — Accuracy, Precision, Recall, F1-Score
Report: PDF output (ReportLab)

Implementation Plan

Data Input Module

The data input module will read family member information from CSV files. Each row will contain one person's document data or one family-related document field. The CSV input will

include NID, Birth Certificate, Academic Certificate, father name, mother name, DOB, and address fields.

Family Document Graph Builder

This module will collect all places where a person's name appears in the family documents. It will connect each person with the related document fields and create the Family Document Graph.

Name Checker and Consensus Engine

The name checker will count all versions of a person's name and select the most common value. It will also classify other values as small mistakes or serious mismatches. Serious mismatches will be marked for senior officer verification.

Date, Age-Gap, and Address Modules

The date module will convert all dates into one standard format. The age-gap module will check whether the parents are older than the children by a proper number of years. The address module will compare Division, District, Upazila, Union, City Corporation, and Pourosova names with a fixed spelling list.

Report Generator

The report generator will show the most common name, mismatch type, DOB result, age-gap result, address spelling suggestion, and final fix suggestions.

Results and Discussion Plan

Evaluation Setup

The proposed method will be tested on the 50 to 100 family CSV test dataset. The old manual correction process will be used as the baseline. The same dataset will be used for both methods.

Quantitative Results

The result table will compare the old method and the proposed method using Accuracy, Precision, Recall, and F1-Score. The evaluation will also count how many name errors, DOB errors, age-gap errors, and address spelling errors are found by each method.

Method	Accuracy	Precision	Recall	F1-Score
Old Manual Method	To be tested	To be tested	To be tested	To be tested
Proposed Method	To be tested	To be tested	To be tested	To be tested

Qualitative Discussion

The discussion will explain which problems the proposed method finds that the old manual method misses. It will also discuss how the mismatch report can help users and officers before a correction application is submitted.

Limitations

This research may have some limitations. The dataset may be small and partly synthetic. The address checker depends on the quality of the fixed spelling list. Manual labeling may also create errors if the correct value is not marked carefully.

Project Timeline

The project runs over two semesters. The first semester is for planning and learning. The second semester is for building, testing, and finishing the project.

Semester 1 (CSE 4000A — 1 Credit)

Phase	Task	Time
Phase 1	Topic Study and Literature Review	Week 1–3
Phase 2	Database and System Design	Week 4–6
Phase 3	Collect and Prepare the CSV Test Dataset	Week 7–9
Phase 4	Proposal Writing and Submission	Week 10–12

Semester 2 (CSE 4000B — 3 Credits) — Main Work

Phase	Task	Time
Phase 5	Build the Old Method (Single Correction Application)	Week 1–2
Phase 6	Build the Family Document Graph and Counting Engine	Week 3–6
Phase 7	Build Name, Date, Age Gap, and Address Checkers	Week 7–9
Phase 8	Build Web Interface and Report Output	Week 10–11
Phase 9	Test Both Methods and Compare Results	Week 12–13
Phase 10	Final Report Writing and Defense Practice	Week 14–16

Conclusion

This project gives a new way to find and fix name problems in Bangladesh civil documents. Instead of handling only one correction application at a time, the system finds every version of a person's name from all family documents, counts which version appears most, and uses that as the correct value. The project tests this new method against the old method to show which one works better. This project works on a real problem that affects many people in Bangladesh. It covers Database Design, Software Engineering, Artificial Intelligence, and Data Science. For this reason, it is a good fit for the Final Year Design Project of the B.Sc. in CSE program at Premier University Chittagong.

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Appendices

Appendix A: Sample Dataset Fields

The CSV dataset will include Family ID, Person ID, Role, NID Name, Birth Certificate Name, Academic Certificate Name, Father Name, Mother Name, DOB, Division, District, Upazila, Union, City Corporation or Pourosova, Address details, and ground-truth correction labels.

Appendix B: Sample Location Master List

The address checker will use a fixed spelling list for Bangladeshi location names. Example entries include Division, District, Upazila, Union, City Corporation, and Pourosova names.

Signature of Group Leader

Name: _____

Date: _____

Signature of Supervisor

Name: _____

Date: _____